CCNA - WAN Technologies

❖ Introduction to WAN Technologies : Definition of WAN, Types of WAN Connections, WAN Devices: CSU/DSU, WAN Aggregators.

➢ Definition of WAN

WAN (Wide Area Network) is a computer network that connects devices, offices, branches, data centers, or users over a large geographical area such as cities, states, countries, or even continents.

WANs are commonly used by organizations to connect branch offices and remote users. Examples of WAN technologies include **leased lines, DSL, MPLS, broadband, 4G/5G, and satellite connections**.

➢ Types of WAN Connections

Some common WAN connection types are:

- **Leased Line:** A dedicated point-to-point connection provided by a telecommunications service provider.
- **DSL:** Uses telephone lines to provide Internet/WAN connectivity.
- **MPLS:** A provider-managed WAN technology that uses labels to efficiently forward traffic between different sites.
- **Broadband:** Uses technologies such as fiber, cable, or other high-speed Internet connections.
- **Wireless WAN:** Uses cellular networks such as 4G and 5G.

➢ WAN Devices

CSU/DSU: A CSU/DSU is used with traditional digital WAN circuits such as T1/E1. The CSU connects the customer equipment to the digital circuit, while the DSU adapts the router's data

interface to the transmission facility and handles functions such as signal timing.

WAN Aggregator: A WAN aggregator is an enterprise networking device that can combine or aggregate multiple WAN connections and services. Cisco's ASR 1000 Series, for example, is specifically designed to aggregate multiple WAN connections and network services.

1. **Create a table comparing at least three types of WAN connections (such as leased line, DSL, and MPLS) with columns for speed, cost, and typical use cases.**

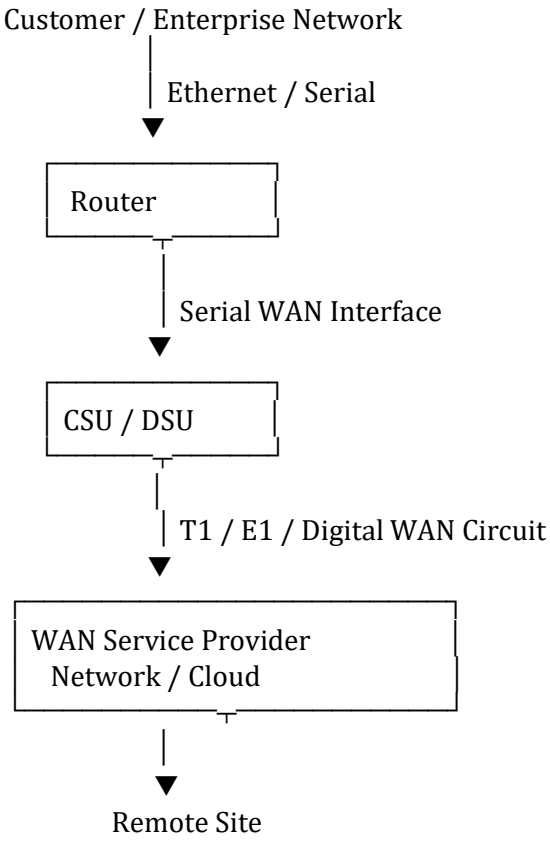
ANS.

WAN Connection	Typical Speed	Cost	Typical Use Cases
Leased Line	From Mbps to multi-Gbps, depending on provider	**High**	Banks, enterprises, data centers, dedicated branch-to-branch connectivity
DSL	Usually from a few Mbps to hundreds of Mbps, depending on technology/provider	**Low**	Small offices, home offices, small businesses
MPLS	From Mbps to multi-Gbps, depending on provider and service	**High**	Large enterprises with multiple branches, managed WAN, business-critical applications
Broadband/Fiber	Usually hundreds of Mbps to multiple Gbps	**Low to Medium**	Internet access, branch offices, cloud/SaaS access
4G/5G WAN	From Mbps to potentially hundreds of Mbps or more	**Medium**	Backup WAN, remote locations, mobile offices

2. Draw a labeled diagram (on paper or using any drawing tool) showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2-3 sentences.

ANS.

- **Labeled Diagram**



- **Role of the Router**

  The **router** connects the organization's internal LAN to the WAN. It determines where IP packets should be forwarded and sends WAN traffic toward the service provider through the WAN interface.

- **Role of the CSU/DSU**

  The **CSU/DSU** provides the interface between the customer's router and a digital WAN circuit. The DSU adapts the router's data interface to the transmission facility and provides timing functions, while the CSU provides the digital line interface toward the telecommunications service.

- **Role of the WAN Service Provider**

  The **WAN service provider** supplies and manages the communication circuit that carries data between geographically separated locations. For traditional leased digital services, the provider's network connects the customer site with the remote site or destination network.

3. List three real-world examples of popular apps or services (like IRCTC, WhatsApp, or Zomato) that rely on WAN connections, and briefly describe how WAN enables their functionality.

ANS.

Application/Service	How WAN Enables Its Functionality
IRCTC	Users from different locations can access IRCTC's online railway ticketing services through the Internet. IRCTC states that its website provides Internet-based transactions with the Indian Railway Passenger Reservation System (PRS). (IRCTC Contents)
WhatsApp	WAN/Internet connectivity allows users in different locations and countries to exchange messages, photos, documents, and make voice/video calls. WhatsApp's own documentation notes that an Internet connection is required for sending and receiving chats and media. (WhatsApp.com)
Zomato	WAN/Internet connectivity allows customers to access the service from different locations, communicate with restaurants and delivery services, place orders, and receive updates through online services.

4. Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device.

ANS.

- **Example: Cisco ASR 1001-X / ASR 1000 Series**

  A suitable real-world example of an enterprise WAN aggregation device is the **Cisco ASR 1000 Series Aggregation Services Router**. Cisco describes the ASR 1000 Series as a platform that **aggregates multiple WAN connections and network services** and supports high-speed WAN aggregation.



- **Two Key Features**

1. **WAN Aggregation:**

   The ASR 1000 Series can aggregate multiple WAN connections and network services, making it suitable for enterprise WAN-edge and aggregation deployments. Cisco's current datasheet describes line-rate forwarding capabilities across the product family from **2.5 to 200 Gbps**, depending on model and configuration.

2. **Security and Advanced WAN Services:**

   The platform supports services such as **encryption, traffic management, firewall/NAT capabilities, and SD-WAN functions**, depending on the model and software configuration. Cisco also documents the ASR 1000 Series for SD-WAN aggregation-headend deployments.

   Official Image/Product Page

   Cisco: **ASR 1000 Series Aggregation Services Routers**